

An illustration of the quantitative problem of the multiple worlds interpretation of quantum mechanics and the motivation for outcome counting

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ABSTRACT. The “quantitative problem” of the multiple worlds interpretation (MWI) of quantum mechanics is to justify the interpretation of the Born rule measure $|a_n|^2$ – the squared norm of the amplitude associated with the n^{th} out of N possible results – as a probability. The essential difficulty is that the basic framework of the MWI would seem to suggest an alternative probability rule, outcome counting, which is that each separate outcome should be equally likely. The purpose of this paper is to illustrate the aesthetic advantages of outcome counting over the Born rule, and to argue that outcome counting is the most “natural” definition of probability in the MWI.¹

RÉSUMÉ. Le le problème quantitatif de l'interprétation des mondes multiples sera obligé à justifier l'interprétation de la mesure de règle de Born $|a_n|^2$ – la norme carrée de l'amplitude associée avec le n^{th} de N les résultats possibles – comme une probabilité. La difficulté essentielle est que le cadre fondamental de l'interprétation des mondes multiples semblerait suggérer une règle de probabilité alternative, le calcul d'issue, qui est que chaque issue séparée doit être également probable. Le but de ce papier sera obligé à illustrer les avantages esthétiques de calcul d'issue par-dessus la règle de Born, et disputer ce calcul d'issue est la définition la plus naturelle de probabilité dans l'interprétation des mondes multiples.

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¹ Archived earlier version of this article may be found in [1].

1 Introduction

In the multiple worlds interpretation (MWI) of quantum mechanics (originally put forth by Everett as the “relative state formulation” [2]), the probability of obtaining a particular experimental outcome is prescribed by the Born rule. According to the Born rule, a probability measure $m_n = |a_n|^2$ is assigned to each of the N experimental results associated with a measurement, with a_n being the complex coefficient of the n^{th} result as calculated by the Schrödinger equation. From an empiric perspective, the success of the Born rule is undisputed.

From an ontologic perspective, the Born rule is not so successful. Perhaps its greatest ontological weakness is what has become known as the “quantitative problem.” Graham, many years ago, expressed the essential difficulty: “it is extremely difficult to see what significance such a measure [the Born rule measure] can have when its implications are completely contradicted by a simple count of the worlds involved, worlds that Everett’s own work assures us must all be on the same footing” [3]. From this perspective, the essential ontology of the MWI would seem to suggest an alternative probability rule, which is that each possible experimental outcome should be equally likely. This alternative probability rule has gone by various names, including “outcome counting” [4] (which will be the term used in this paper), “egalitarianism” [5], or the “alternate projection postulate” [6]. The difficulty with outcome counting, of course, is immediately obvious: if one simply replaces the Born rule with outcome counting, the predictions are *wrong*. Any attempt to fix the model through additional modification, so that predictions are correct², would likely – or so the argument goes – be so difficult, or ugly, or otherwise unappealing, that there is no use even making the attempt. Better it would be not to worry so much about ontology.

Nevertheless, for those of us who worry about ontology, this situation is unsatisfying. It feels as if there is a form of double-speak behind the interpretation of the Born rule, whereby each world is “equally real” – except that some worlds are “more real” (more probable) than others.

The purpose of this paper is to *illustrate* why it is that this state of affairs is unsatisfactory, the purpose being to motivate the construction

²In such a model, we would expect outcome counting, as the fundamental probability rule, to operate at the fine-grained level, and the Born rule to operate at a coarser-grained level. The Born rule would not be *wrong*, per se; but its implementation would be such that it no longer plays the role that it does in our understanding of “what probability really is.”

of a model based on outcome counting. This will be performed through the analysis of an experiment commonly encountered in quantum mechanics: a spin measurement. This analysis will define and make use of an axiom labeled here the “probability criterion,” which – loosely (and somewhat facetiously) stated – is that it is better to be more frequently right than more frequently wrong. The argument recapitulates but also expands upon a similar argument by Graham [3]. For a more abstract philosophic analysis, the reader is referred to a growing literature [7] [5] [8] [9] [10].

2 A simple experiment

Consider an experiment in which M identically prepared spin-1/2 particles are prepared, with their spins being measured sequentially. Using the Born rule, each particle is predicted to be observed to be spin up or down with probabilities m_{up} or m_{down} , respectively. Since each individual spin measurement produces two separate branches (worlds), there will be a total of 2^M worlds at the end of the M measurements. In each of these worlds, upon the completion of the M measurements, the observer is imagined to calculate the frequency p_{up} with which the particles were observed to be in the spin up state. In other words, m_{up} is what the theory predicts, and p_{up} is what is actually observed (with m_{up} being calculated once prior to experiment, and p_{up} being calculated separately in each of the 2^M worlds, after completion of the experiment). From a practical perspective, the prediction is tested by comparing the prediction m_{up} with the observation p_{up} (and likewise for spin down), using as large a value for M as is practically feasible. In particular, it is hoped that as $M \rightarrow \infty$, the physical measure of the number of worlds in which the Born rule appears to be false – that is, in which p_{up} deviates from m_{up} by an arbitrarily chosen (small) number δ – approaches zero.

The notion of testing a “probability rule” by comparing the predicted frequency to the observed frequency may be stated more generally for any arbitrary quantum mechanical experiment, using the following definitions:

Definition 1 *Experiments M . M is the number of times that the experiment is run.*

Definition 2 *Outcomes N . N is the number of mutually exclusive possible outcomes for one experimental trial.*

Note the assumption that the spectrum of experimental outcomes is *discrete*. This assumption may be made without loss of generality, and is necessary for the purpose of defining a *measure* over the number of worlds in which a particular outcome is observed. A quantum mechanical experiment with a continuous spectrum of outcomes, such as a position measurement, should more properly be conceptualized as a theoretical (and perhaps unattainable) limit as the number of discrete elements of the position measurement apparatus approaches infinity.

Definition 3 *Observed probability measure p_n . For any individual world (of which there are N^M), $P_n \in [0, M]$ is the number times that the n^{th} outcome was observed, with $p_n = P_n/M \in [0, 1]$ being the frequency of this outcome among the M measurements.*

Definition 4 *Predicted probability measure m_n . m_n is the predicted probability associated with the n^{th} outcome, $n \in [1, \dots, N]$.*

Note that p_n is an attribute of an individual world, whereas m_n is a predicted quantity that is independent of any individual experimental result. The expression for m_n will depend on which probability rule we choose; e.g., the Born rule $m_n = m_n^{\text{Born}} = |a_n|^2$, outcome counting $m_n = m_n^{\text{OC}} = 1/N$, or some other rule.

Definition 5 *Error ϵ_n . The difference between the predicted probability measure m_n and the observed quantity p_n will be referred to as the “error” ϵ_n , that is, $\epsilon_n = |p_n - m_n|$.*

Definition 6 *Measure of belief F_n . $F_n \in [0, 1]$ is the proportion of the N^M worlds in which the observer concludes that the probability measure is valid, as determined by the error being less than or equal to an arbitrarily chosen cutoff, $\epsilon_n \leq \delta$.*

Given these definitions, the probability criterion may be stated in the following manner:

Axiom 1 *Probability criterion. For any arbitrarily chosen δ , $\lim_{M \rightarrow \infty} F_n = 1$.*

The probability criterion is essentially a mathematical statement of the notion that “*most*” of the worlds will produce an observer who concludes that the theoretical prediction is correct, to within some arbitrarily chosen cutoff. In a sense, the probability criterion may be interpreted as a *definition* of the very notion of probability: it is a requirement that must be met by the probability measure m_n . It therefore may come as a surprise that the probability criterion is not generally met if m_n is calculated by the Born rule, $m_n = m_n^{Born} = |a_n|^2$. That is to say that in *most* of the N^M worlds, the observer will conclude that the Born rule is false! The probability criterion *is* met, however, if m_n is calculated by outcome counting, $m_n = m_n^{OC} = 1/N$.

By way of illustration, suppose $m_{up} = 0.9$, $m_{down} = 0.1$, $\delta = 0.1$ and $M = 100$. These 100 measurements result in 2^{100} worlds, and it is asked: in what proportion of these worlds does the observer find that p_{up} falls within the interval $m_{up}^{Born} \pm \delta$? A quick calculation (using the formulae derived in the next section) shows that the answer is a miniscule 5.58×10^{-8} percent. Stated differently: the proportion of worlds in which the observer finds that the Born rule is *wrong* is a rather large 99.999999442 percent. If we are to accept the Born rule, then we must accept that these worlds, somehow, just don’t matter.

In contrast, the probability criterion *is* met if the probability measure is calculated using outcome counting, $m_{up}^{OC} = 1/N$. In this case, $m_{up}^{OC} = 1/2$, and the proportion of worlds in which the observer finds that p_{up} falls within the interval $m_{up}^{OC} \pm \delta$ is a much larger 96.4 percent. It is simple to see that for any arbitrary δ , as M approaches ∞ , the former value approaches zero, and the latter approaches 100 percent.

3 Generalization

It is not so difficult to see that outcome counting is a general solution to the probability criterion, and so is *generally* immune to the above difficulty. Assume the definitions above. It is further assumed that m_n can be expressed as a function of N . (Compare this to Everett’s assumption [2], discussed in [6], that m_n can be expressed as a function of a_n .) The goal is to find a general expression that tells us whether a given m_n does or does not satisfy the probability criterion.

The number of ways to make P_n observations of the n^{th} outcome is calculated to be $\frac{(N-1)^{(M-P_n)} M!}{P_n! (M-P_n)!}$, as follows. There is only one way to get the n^{th} result on P_n out of P_n trials. Next, the number of ways to get

anything other than the n^{th} result on $M - P_n$ out of $M - P_n$ trials is $(N-1)^{M-P_n}$. Next, the number of ways to mix an ordered sequence with P_n elements and an ordered sequence of $M - P_n$ elements, i.e. the number of ways of distributing P_n elements over M elements is M choose P_n , i.e. $\frac{M!}{P_n!(M-P_n)!}$. Multiplying these expressions yields: $\frac{(N-1)^{(M-P_n)}M!}{P_n!(M-P_n)!}$. Dividing this expression by the total number of worlds N^M yields the proportion $f(p_n) \in [0, 1]$ of such worlds: $f(p_n) = \frac{(N-1)^{(M-P_n)}M!}{P_n!(M-P_n)!N^M}$. The integrated proportion $F(m_n, \delta) \in [0, 1]$ of worlds in which the observed frequency p_n is close to the predicted frequency m_n , i.e. falls anywhere within the closed interval $m_n \pm \delta$, i.e. falls within $[m_n - \delta, m_n + \delta]$, is calculated as the sum: $F(m_n, \delta) = \sum_{p_n=m_n-\delta}^{p_n=m_n+\delta} f(p_n)$. The probability criterion states that for any δ , $\lim_{M \rightarrow \infty} F(m_n, \delta) = 1$. By setting $m_n = 1/N$ in the above expression, it is readily seen that the outcome counting rule does in fact satisfy this equation. Computerized numeric calculations confirms this solution. Therefore, it can be concluded that *outcome counting is a general solution to the probability criterion*.

The essential similarity between outcome counting and the probability criterion is that in both cases, each of the N possible “branches” associated with a single measurement – or equivalently, each of the N^M distinct worlds resulting from M measurements – is considered to be, ontologically speaking, on an “equal footing.” The Born rule, on the other hand, does not seem to offer a clear ontological picture of the “reality” of alternate worlds. The fact that outcome counting, but not the Born rule, is a solution to the probability criterion is precisely what makes it the more natural choice for a probability rule.

4 Discussion

The purpose of this paper has been to present the aesthetic appeal of outcome counting as an alternative to the Born rule for the calculation of probabilities in the MWI. We have seen that if a theory is based on outcome counting, then the measure of worlds in which prediction matches observation approaches unity. On the other hand, if a theory is based on the Born rule, then the measure of worlds in which prediction does *not* match observation approaches unity. One must merely decide which of these two options is the more desirable.

Whether or not this state of affairs is troubling is, perhaps, no more than a matter of taste. As it now stands, many people, including Everett himself, have dedicated much effort towards the demonstration that the

Born rule is a natural consequence of the essential structure of the MWI – thus rendering outcome counting not even an option, and hence “solving” the quantitative problem. Hartle has contributed to these attempts [11]. More recently, Deutsch-Wallace decision theory [7] [12] has been proposed to demonstrate the inevitability of the Born rule, essentially by deriving it *a priori*. Others, however, have criticized derivations of these types on the grounds that they are based on circular reasoning, because they contain hidden probabilistic assumptions that effectively “sneak” the Born rule into the formalism, so that they in fact assume what they have set out to prove [6] [10] [13]. The ontological status of the probability interpretation of the MWI, therefore, continues to be a subject of controversy.

There is another option: accept outcome counting as fundamental. Quantum mechanics, it may be said, is nothing more and nothing less than a theory about probability. So the *interpretation* of probability could be argued to be the most important interpretational issue of the theory. Given the aesthetic appeal of outcome counting, we might elevate it to the level of a symmetry principle, on a par with, say, the principle of relativity. Obviously, the principle of relativity plays a central role in the construction of the theory of relativity. Could outcome counting play a similarly important role in a theory of nature?

It may come as a surprise that it is even *possible* to construct a version of the MWI that is based on outcome counting *but also succeeds as a description of the laws of nature*. The key is to identify some *additional* modification (besides replacing the Born rule with outcome counting) so that quantum statistics may be restored to the overall programme. There are, in fact, several independently proposed models. One is the “mangled worlds” model [14], according to which the memory of observers in “small” worlds is destroyed or somehow converted to remember events from “large” worlds. (This results in elimination of the 99.9999999442 percent of worlds, discussed above, in which the Born rule appears to be false.) A second proposal [4] assumes nonlinearity in the time-dependent evolution of the observer state, as well as an exponential time dependence for the nonlinear effects. A third model (the work of the author, to be published separately) stipulates that our method of counting worlds has been incorrect all along, and suggests a novel method for enumerating the number of distinct physical states (outcomes) associated with a given experiment.³

³Graham [3] introduced his own model; however, I have failed to understand how

The main point of this paper is to suggest that these efforts to base a theory on outcome counting have been too few and far between. Instead of taking an empirically successful rule – the Born rule – and trying to demonstrate that “it could be no other way,” we should at least consider a different approach: to assume outcome counting as fundamental, and see where it leads us. It could just be that this is – or will help to identify – one small, yet essential ingredient to a theory of quantum gravity.

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